

# Medical Statistics - Answers lab 5

## Introduction

This answer page gives one complete set of answers for lab 5. The same numerical results should be obtained when the analyses are carried out in R or SPSS.

The analyses use the Cleveland heart disease dataset and address the following research questions:

- Does maximum heart rate achieved differ across chest pain types?
- Is the presence of heart disease associated with sex?
- Is the presence of heart disease associated with fasting blood sugar?

## Exercise 1: Download and open the dataset

The dataset contains 303 patients and 14 variables.

```
library(haven)
library(knitr)
library(tableone)
library(car)
library(dunn.test)

heart_data <- as.data.frame(read_sav("datasets/heart_disease_cleveland.sav"))

# Convert labelled categorical variables to factors.
labelled_vars <- vapply(heart_data, haven::is_labelled, logical(1))
heart_data[labelled_vars] <- lapply(heart_data[labelled_vars], haven::as_factor)

dim(heart_data)
```

```
[1] 303 14
```

```
names(heart_data)
```

```
[1] "age"      "sex"      "cp"      "trestbps" "chol"     "fbs"
[7] "restecg"  "thalach"  "exang"    "oldpeak"  "slope"    "ca"
[13] "thal"     "target"
```

The variables `sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `thal`, and `target` are categorical. The variable `ca` is also best treated as categorical for the baseline table, because it records the number of major vessels from 0 to 3.

```
heart_data$ca <- factor(heart_data$ca)
```

## Exercise 2: Create a baseline characteristics table

```
continuous_vars <- c("age", "trestbps", "chol", "thalach", "oldpeak")
categorical_vars <- c("sex", "cp", "fbs", "restecg", "exang", "slope", "ca", "thal")
all_vars <- c(continuous_vars, categorical_vars)

table1 <- CreateTableOne(
  vars = all_vars,
  data = heart_data,
  factorVars = categorical_vars
)

kable(
  print(
    table1,
    nonnormal = "oldpeak",
    printToggle = FALSE
  )
)
```

|                      | Overall        |
|----------------------|----------------|
| n                    | 303            |
| age (mean (SD))      | 54.44 (9.04)   |
| trestbps (mean (SD)) | 131.69 (17.60) |
| chol (mean (SD))     | 246.69 (51.78) |
| thalach (mean (SD))  | 149.61 (22.88) |

|                              | Overall           |
|------------------------------|-------------------|
| oldpeak (median [IQR])       | 0.80 [0.00, 1.60] |
| sex = Male (%)               | 206 (68.0)        |
| cp (%)                       |                   |
| Typical angina               | 23 ( 7.6)         |
| Atypical angina              | 50 (16.5)         |
| Non-anginal pain             | 86 (28.4)         |
| Asymptomatic                 | 144 (47.5)        |
| fbs = True (%)               | 45 (14.9)         |
| restecg (%)                  |                   |
| Normal                       | 151 (49.8)        |
| ST-T wave abnormality        | 4 ( 1.3)          |
| Left ventricular hypertrophy | 148 (48.8)        |
| exang = Yes (%)              | 99 (32.7)         |
| slope (%)                    |                   |
| Upsloping                    | 142 (46.9)        |
| Flat                         | 140 (46.2)        |
| Downsloping                  | 21 ( 6.9)         |
| ca (%)                       |                   |
| 0                            | 176 (58.9)        |
| 1                            | 65 (21.7)         |
| 2                            | 38 (12.7)         |
| 3                            | 20 ( 6.7)         |
| thal (%)                     |                   |
| Normal                       | 166 (54.8)        |
| Fixed defect                 | 20 ( 6.6)         |
| Reversible defect            | 117 (38.6)        |

A suitable baseline table gives one summary format per variable. In the table above, most continuous variables are summarized using mean and standard deviation, `oldpeak` is summarized using median and interquartile range because it is clearly skewed, and categorical variables are summarized using counts and percentages.

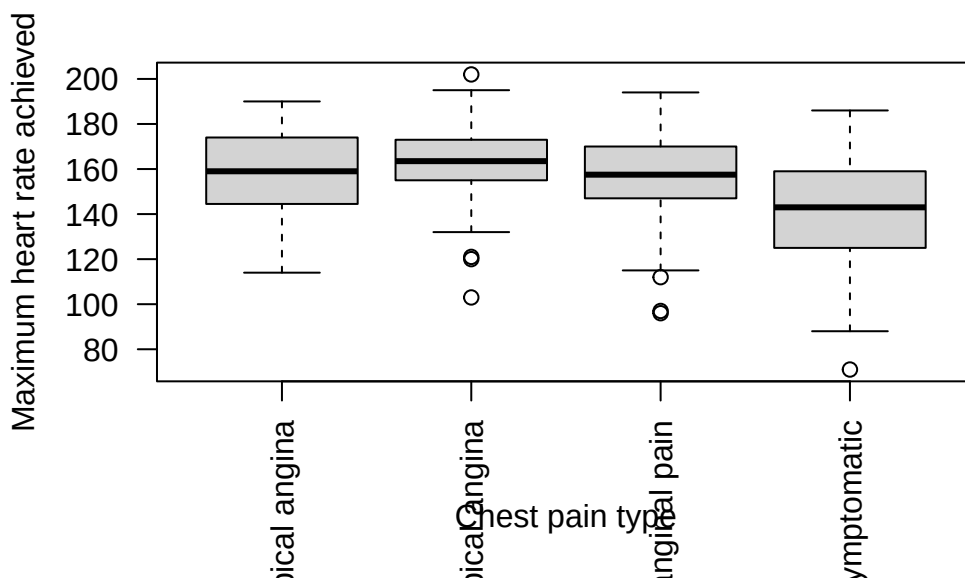
### Exercise 3: Analyze maximum heart rate across chest pain types

Research question: *Does maximum heart rate achieved differ across chest pain types?*

The outcome is `thalach`, and the grouping variable is `cp`.

## Descriptive results

```
boxplot(  
  thalach ~ cp,  
  data = heart_data,  
  xlab = "Chest pain type",  
  ylab = "Maximum heart rate achieved",  
  las = 2  
)
```



```
for (group in levels(heart_data$cp)) {  
  group_data <- heart_data$thalach[heart_data$cp == group]  
  n <- length(group_data)  
  mean_thalach <- mean(group_data)  
  se_thalach <- sd(group_data) / sqrt(n)  
  ci_thalach <- mean_thalach + c(-1, 1) * qt(0.975, df = n - 1) * se_thalach  
  
  cat(  
    group,  
    ": n =", n,  
    ", mean =", round(mean_thalach, 1),  
    ", 95% CI = [", round(ci_thalach[1], 1), ", ", round(ci_thalach[2], 1), "]\n",  
    file = "results.txt", append = TRUE)
```

```

    sep = ""
  )
}

```

Typical angina: n =23, mean =156, 95% CI = [146.9, 165]  
 Atypical angina: n =50, mean =162.4, 95% CI = [157, 167.8]  
 Non-anginal pain: n =86, mean =155.4, 95% CI = [151.3, 159.5]  
 Asymptomatic: n =144, mean =140.7, 95% CI = [136.9, 144.5]

The mean maximum heart rate was lowest in the asymptomatic chest pain group.

## One-way ANOVA

```

anova_model <- aov(thalach ~ cp, data = heart_data)
summary(anova_model)

```

|           | Df  | Sum Sq | Mean Sq | F value | Pr(>F)       |
|-----------|-----|--------|---------|---------|--------------|
| cp        | 3   | 23503  | 7834    | 17.41   | 1.91e-10 *** |
| Residuals | 299 | 134523 | 450     |         |              |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

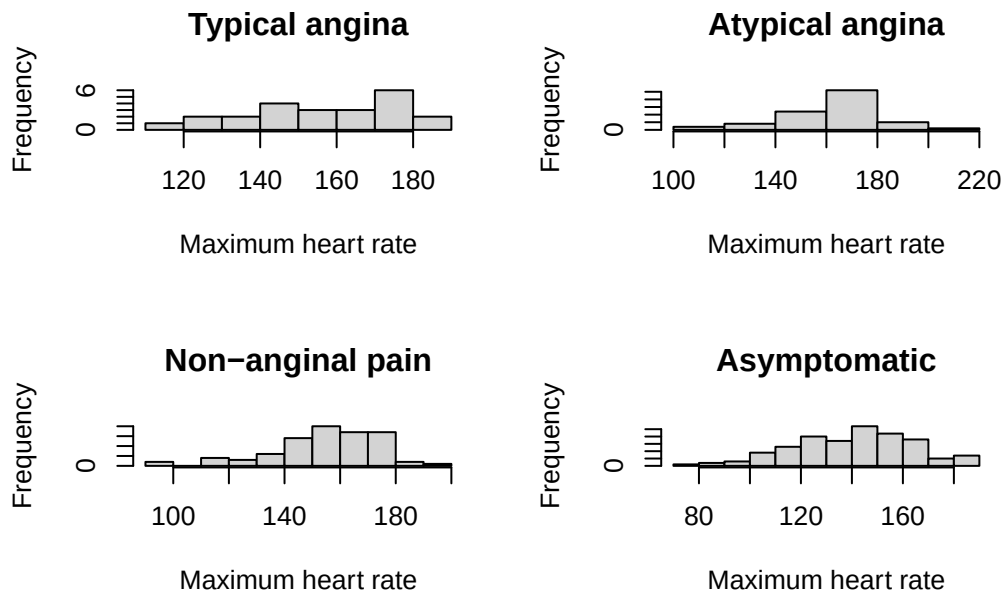
The one-way ANOVA is statistically significant,  $F(3, 299) = 17.41$ ,  $p < 0.001$ . This indicates that maximum heart rate achieved differs across chest pain types.

## Assumption checks

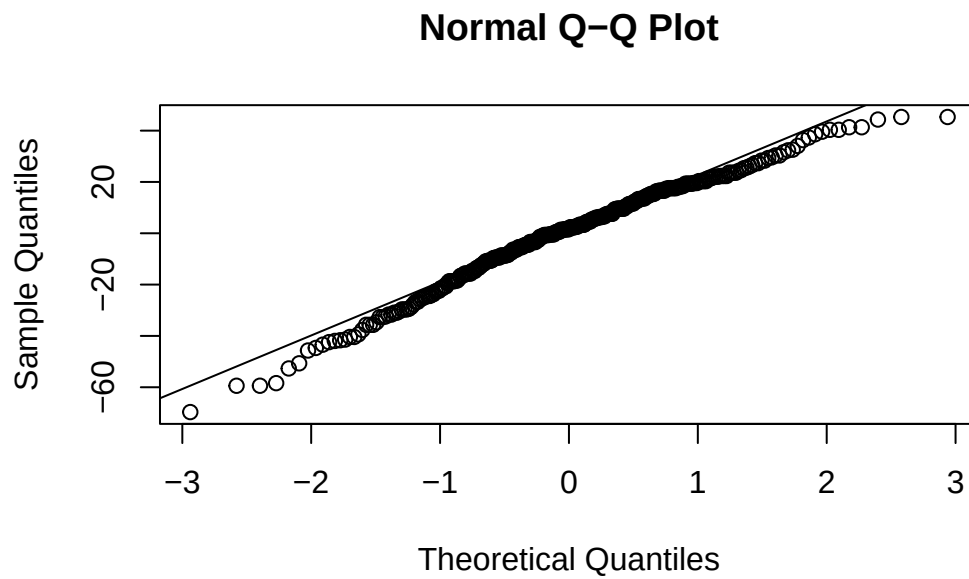
```

par(mfrow = c(2, 2))
hist(heart_data$thalach[heart_data$cp == "Typical angina"],
     main = "Typical angina", xlab = "Maximum heart rate")
hist(heart_data$thalach[heart_data$cp == "Atypical angina"],
     main = "Atypical angina", xlab = "Maximum heart rate")
hist(heart_data$thalach[heart_data$cp == "Non-anginal pain"],
     main = "Non-anginal pain", xlab = "Maximum heart rate")
hist(heart_data$thalach[heart_data$cp == "Asymptomatic"],
     main = "Asymptomatic", xlab = "Maximum heart rate")

```



```
par(mfrow = c(1, 1))
qqnorm(residuals(anova_model))
qqline(residuals(anova_model))
```



```
tapply(heart_data$thalach, heart_data$cp, sd)
```

| Typical angina | Atypical angina | Non-anginal pain | Asymptomatic |
|----------------|-----------------|------------------|--------------|
| 20.89906       | 18.98882        | 19.31679         | 22.98210     |

```
leveneTest(thalach ~ cp, data = heart_data)
```

Levene's Test for Homogeneity of Variance (center = median)

|       | Df  | F value | Pr(>F)  |
|-------|-----|---------|---------|
| group | 3   | 2.4694  | 0.06209 |
|       | 299 |         |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

The histograms and Q-Q plot can be used to assess whether the distributions are roughly compatible with ANOVA. There is no need to base this decision on a formal test of normality; with these sample sizes such tests are often not helpful for deciding which analysis to use. The group standard deviations are of a similar order of magnitude, and Levene's test does not show evidence against the equal variances assumption.

## Post-hoc comparisons

```
pairwise.t.test(
  heart_data$thalach,
  heart_data$cp,
  p.adjust.method = "bonferroni",
  pool.sd = TRUE
)
```

Pairwise comparisons using t tests with pooled SD

data: heart\_data\$thalach and heart\_data\$cp

|                  | Typical angina | Atypical angina | Non-anginal pain |
|------------------|----------------|-----------------|------------------|
| Atypical angina  | 1.0000         | -               | -                |
| Non-anginal pain | 1.0000         | 0.3840          | -                |
| Asymptomatic     | 0.0089         | 8.8e-09         | 3.7e-06          |

P value adjustment method: bonferroni

Using Bonferroni-adjusted pairwise t-tests, the asymptomatic chest pain group has a significantly lower maximum heart rate than the typical angina group, the atypical angina group, and the non-anginal pain group. The other pairwise comparisons are not statistically significant after Bonferroni adjustment.

### Kruskal-Wallis sensitivity analysis

```
kruskal.test(thalach ~ cp, data = heart_data)
```

Kruskal-Wallis rank sum test

data: thalach by cp

Kruskal-Wallis chi-squared = 47.315, df = 3, p-value = 2.979e-10

```
dunn.test(
  heart_data$thalach,
  heart_data$cp,
  method = "bonferroni",
  altp = TRUE
)
```

Kruskal-Wallis rank sum test

data: x and group

Kruskal-Wallis chi-squared = 47.3151, df = 3, p-value = 0

#### Comparison of x by group (Bonferroni)

| Col Mean- |           |          |           |
|-----------|-----------|----------|-----------|
| Row Mean  | Asymptom  | Atypical | Non-angi  |
| -----+    | -----     |          |           |
| Atypical  | -5.970501 |          |           |
|           | 0.0000*   |          |           |
|           |           |          |           |
| Non-angi  | -4.836917 | 1.804218 |           |
|           | 0.0000*   | 0.4272   |           |
|           |           |          |           |
| Typical   | -2.983404 | 1.230885 | -0.045777 |
|           | 0.0171*   | 1.0000   | 1.0000    |



```
alpha = 0.05
Reject Ho if p <= alpha
```

The Kruskal-Wallis test gives the same substantive conclusion,  $\chi^2(3) = 47.32$ ,  $p < 0.001$ . The Bonferroni-adjusted Dunn's test again shows that the asymptomatic group differs from the other three chest pain groups.

#### Exercise 4: Analyze heart disease by sex

Research question: *Is the presence of heart disease associated with sex?*

##### Note

For the 2x2 tables in this lab, `chisq.test(..., correct = FALSE)` is used to report the ordinary Pearson chi-square test. This corresponds to the **Pearson Chi-Square** row in SPSS. If `correct = FALSE` is omitted, R applies Yates' continuity correction by default for 2x2 tables; SPSS reports this separately as **Continuity Correction**.

```
sex_table <- table(heart_data$sex, heart_data$target)
sex_table
```

|        | Heart Disease | No Heart Disease |
|--------|---------------|------------------|
| Female | 25            | 72               |
| Male   | 114           | 92               |

```
round(prop.table(sex_table, margin = 1) * 100, 1)
```

|        | Heart Disease | No Heart Disease |
|--------|---------------|------------------|
| Female | 25.8          | 74.2             |
| Male   | 55.3          | 44.7             |

The row percentages show that the prevalence of heart disease was higher in males than in females.

The expected cell counts are all larger than 5, so the chi-square test is appropriate.

```
sex_chisq <- chisq.test(sex_table, correct = FALSE)
sex_chisq$expected
```

|        | Heart Disease | No Heart Disease |
|--------|---------------|------------------|
| Female | 44.49835      | 52.50165         |
| Male   | 94.50165      | 111.49835        |

```
sex_chisq
```

Pearson's Chi-squared test

```
data: sex_table
X-squared = 23.218, df = 1, p-value = 1.446e-06
```

There is strong evidence of an association between sex and heart disease,  $\chi^2(1) = 23.22$ ,  $p < 0.001$ . The prevalence of heart disease was higher in males (55.3%) than in females (25.8%).

## Exercise 5: Analyze heart disease by fasting blood sugar

Research question: *Is the presence of heart disease associated with fasting blood sugar?*

```
fbs_table <- table(heart_data$fbs, heart_data$target)
fbs_table
```

|       | Heart Disease | No Heart Disease |
|-------|---------------|------------------|
| False | 117           | 141              |
| True  | 22            | 23               |

```
round(prop.table(fbs_table, margin = 1) * 100, 1)
```

|       | Heart Disease | No Heart Disease |
|-------|---------------|------------------|
| False | 45.3          | 54.7             |
| True  | 48.9          | 51.1             |

The row percentages show similar heart disease prevalence in the two fasting blood sugar groups.

The expected cell counts are all larger than 5, so the chi-square test is appropriate.

```
fbs_chisq <- chisq.test(fbs_table, correct = FALSE)
fbs_chisq$expected
```

|       | Heart Disease | No Heart Disease |
|-------|---------------|------------------|
| False | 118.35644     | 139.64356        |
| True  | 20.64356      | 24.35644         |

```
fbs_chisq
```

Pearson's Chi-squared test

```
data: fbs_table
X-squared = 0.19339, df = 1, p-value = 0.6601
```

There is no evidence of an association between fasting blood sugar and heart disease,  $\chi^2(1) = 0.19$ ,  $p = 0.660$ . The prevalence of heart disease was similar for patients with fasting blood sugar at or below 120 mg/dl (45.3%) and patients with fasting blood sugar above 120 mg/dl (48.9%).